

Credit Risk Analysis of a Healthcare Sector Portfolio Using One Factor Model and Monte Carlo Simulation

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1 Introduction

This report evaluates the credit risk of a healthcare-sector loan portfolio using a market-calibrated one-factor model. Monte Carlo simulations and an analytical approximation (Vasicek-style) are applied to estimate key risk metrics: Expected Loss (EL), Value at Risk (VaR), and Expected Shortfall (ES).

2 Data Processing

The dataset included:

- `companies.csv`: company metadata
- `ratings.csv`: credit ratings for each company
- `portfolio2_composition.csv`: portfolio exposures
- `rating_transitions.csv`: transition matrix with PDs
- `healthcare_stock_returns.csv` and `healthcare_index.csv`: daily returns for correlation analysis

2.1 Rating Distribution

The portfolio contains the following credit quality distribution:

Rating	Count
AAA	2
AA	2
A	12
BBB	15
BB	10
B	7
CCC	12

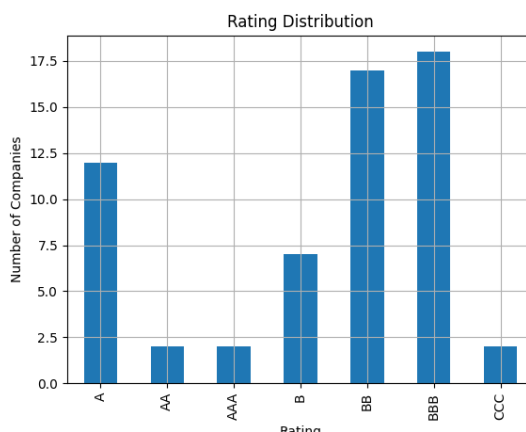


Figure 1: Rating Distribution

2.2 PD Thresholds

Each company's credit rating was mapped to a default probability (PD), which was then transformed into a default threshold using the inverse standard normal CDF $\Phi^{-1}(\text{PD})$.

2.3 Asset Correlations

Correlations between individual stock returns and the healthcare sector index were used to estimate company-specific asset correlations ρ_i .

3 Model Setup: One Factor Model

Credit events are driven by a firm's latent asset return:

$$A_i = \rho_i \cdot F + \sqrt{1 - \rho_i^2} \cdot \epsilon_i$$

- $F \sim \mathcal{N}(0, 1)$: market-wide factor from healthcare index
- $\epsilon_i \sim \mathcal{N}(0, 1)$: firm-specific idiosyncratic noise
- Default occurs if $A_i < D_i = \Phi^{-1}(\text{PD}_i)$

4 Monte Carlo Simulation

Simulation process:

1. $N = 10,000$ scenarios
2. For each simulation:
 - (a) Draw F from $\mathcal{N}(0, 1)$
 - (b) For each firm, simulate ϵ_i
 - (c) Calculate A_i and determine default
 - (d) Aggregate loss as sum of exposures in default

4.1 Results

Monte Carlo Simulation Results (with LGD = 45%)

- **Expected Loss (EL):** \$9,794.63
- **99% Value-at-Risk (VaR):** \$44,807.90
- **99% Expected Shortfall (ES):** \$52,975.66

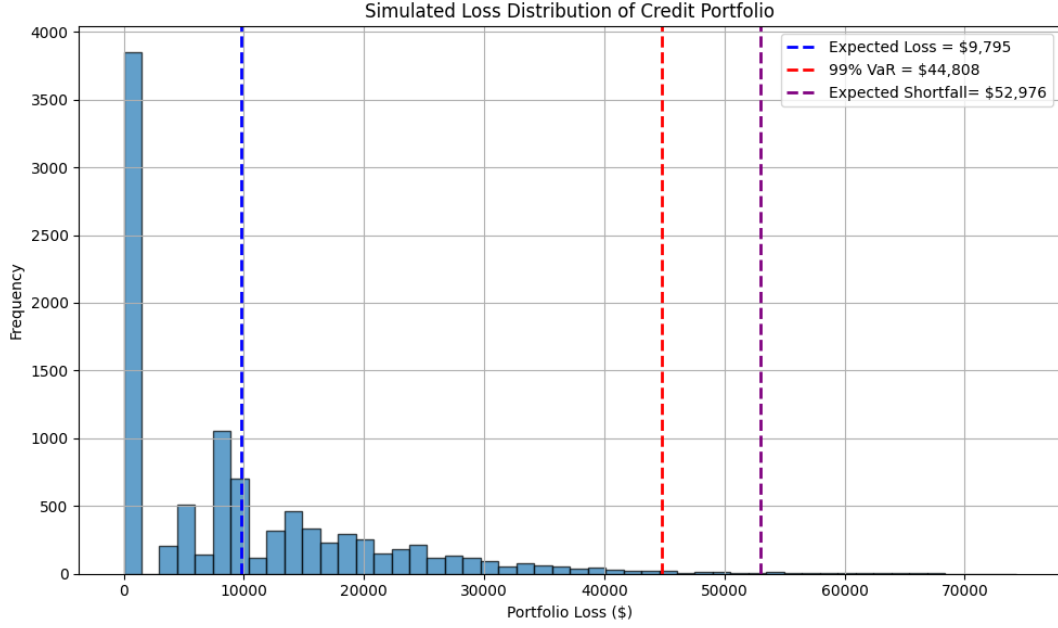


Figure 2: Simulated Portfolio Loss Distribution with Risk Metrics (LGD = 45%)

Key Observations

- The Expected Loss is approximately \$9.8K, reflecting the average loss adjusted for default probabilities.
- The 99% VaR of \$44.8K suggests that losses are unlikely to exceed this threshold more than 1% of the time.
- The 99% Expected Shortfall (ES) is \$52.98K, indicating that if losses exceed the VaR, they average close to \$53K.

Analytical Approximation (Vasicek Model at 99%)

Using a Vasicek-style approximation:

- **Analytical Expected Loss:** \$9,590.08
- **Analytical 99% VaR:** \$10,574.66
- **Estimated Capital Requirement (99% VaR - Expected Loss):** \$984.58
- **Sensitivity Analysis:** The analytical VaR is highly sensitive to the asset correlation assumption (ρ). As ρ increases, systemic risk rises, substantially elevating the capital requirement. This sensitivity is summarized below.

Correlation (ρ)	Expected Loss (\$)	VaR (99%) (\$)	Capital Requirement (\$)
0.1	9,590	16,151	6,561
0.3	9,590	37,516	27,926
0.5	9,590	73,582	63,992
0.7	9,590	132,450	122,860

Table 1: Sensitivity of Analytical 99% VaR to Correlation Assumptions

Metric	Monte Carlo (99%)	Analytical (99%)	Difference
Expected Loss	\$9,795	\$9,590	+2.1%
VaR	\$44,808	\$10,575	+323.1%

Table 2: Monte Carlo vs. Analytical Results (LGD = 45%)

5 Conclusion and Recommendations

Conclusion

- The Monte Carlo simulation effectively captures portfolio heterogeneity, non-linear correlations, and extreme loss scenarios, providing a realistic representation of credit risk.
- Results from the Monte Carlo simulation indicate a relatively low average Expected Loss (EL) of \$9.8K but reveal significant tail risk, with an Expected Shortfall (ES) exceeding \$50K.
- In contrast, the Vasicek-based analytical approximation, while computationally efficient, substantially underestimates tail risk (e.g., a 99% Value at Risk (VaR) of just \$10.6K), highlighting its limitations in capturing complex dependencies.
- The Vasicek model’s output is highly sensitive to the assumed asset correlation. For example, increasing ρ from 0.1 to 0.7 can amplify the capital requirement nearly 19-fold, emphasizing the importance of conservative parameter calibration.

Recommendations

- Reduce exposure to low-rated (B, CCC) or highly market-correlated firms, and rebalance toward higher-quality, lower-risk credits.
- Enhance realism by using stochastic LGD, adopting copula or multi-factor models, and conducting stress tests on market and rating dynamics.
- Base capital buffers on Expected Shortfall and regularly stress test for macroeconomic shocks to capture tail-risk sensitivity.